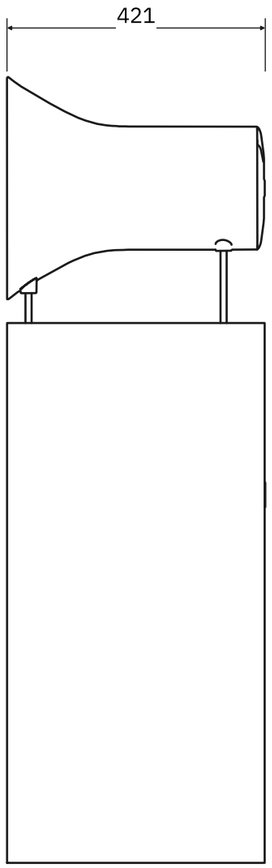
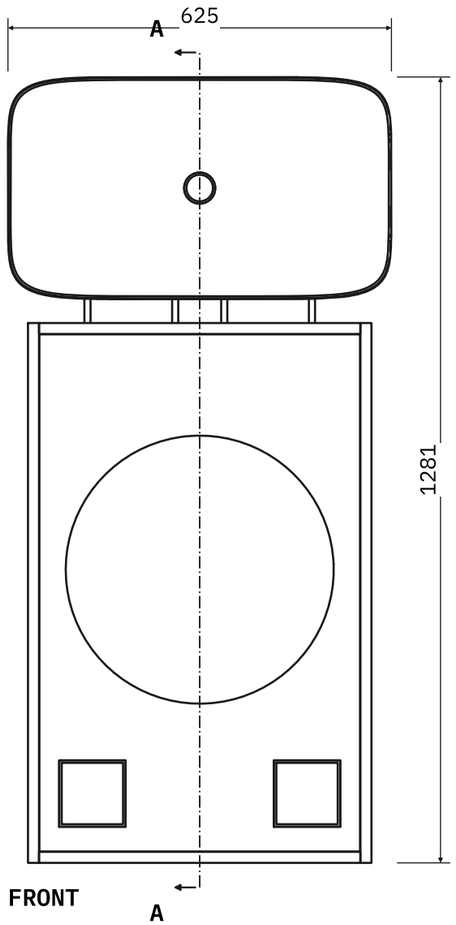
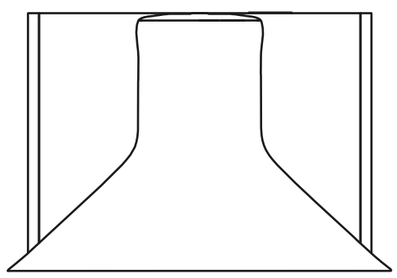
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

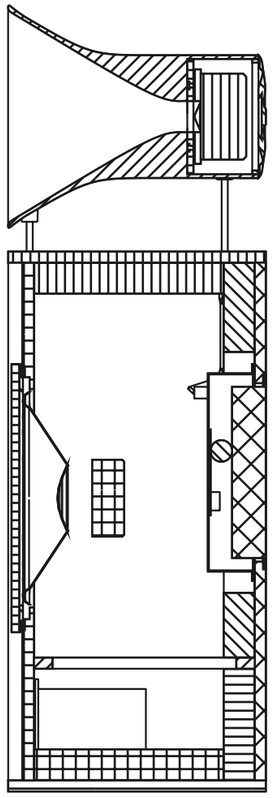
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



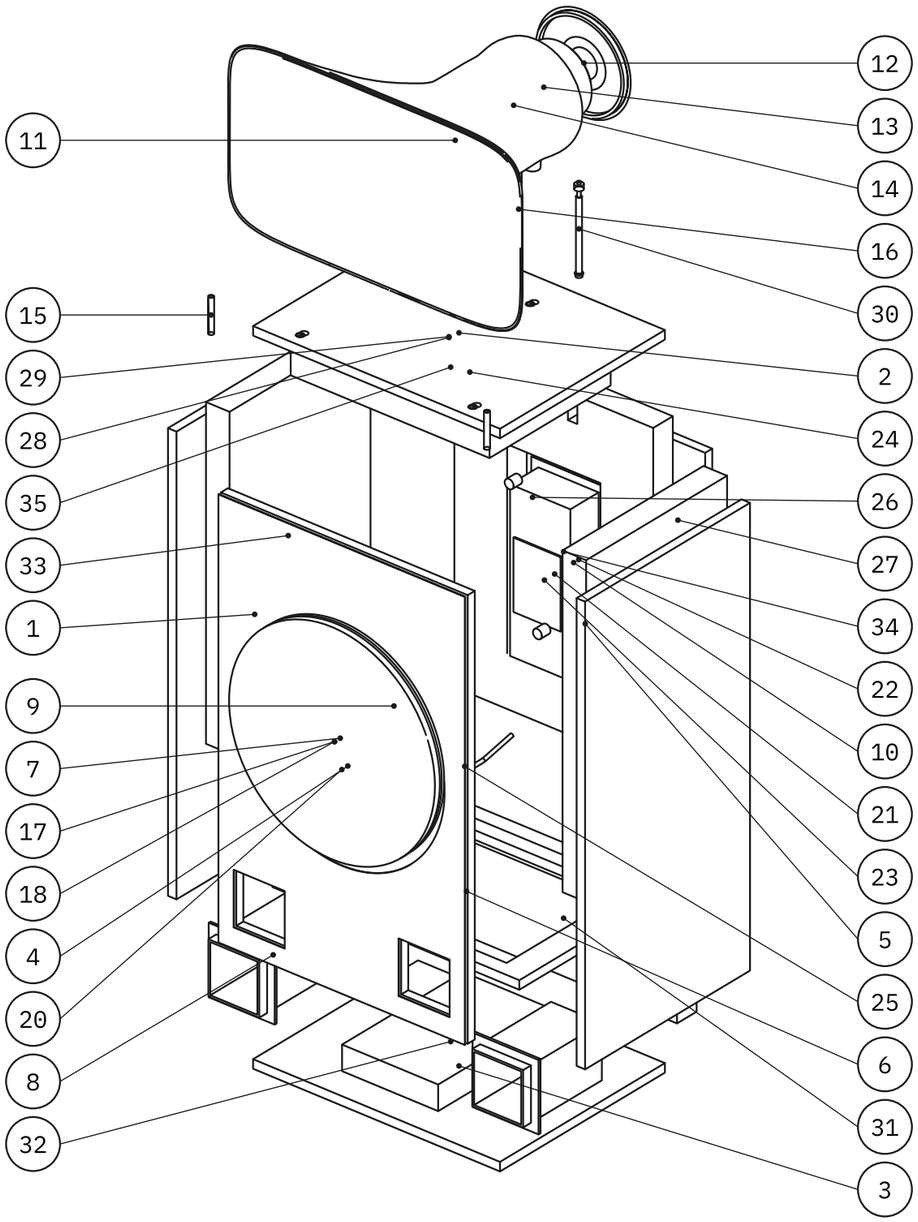
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



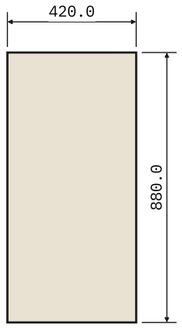
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

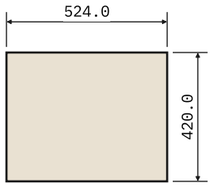
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:20. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



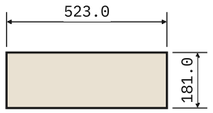
left, right

18 mm baltic-birch, 4 for the set



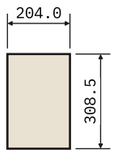
bottom

18 mm baltic-birch, 2 for the set



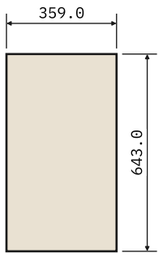
lining-back-1

50 mm polyester-batting, 2 for the set



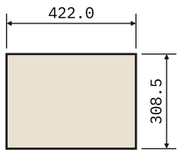
lining-bottom-1

50 mm polyester-batting, 2 for the set



lining-left-2, lining-right-2

50 mm polyester-batting, 4 for the set



lining-top-2

50 mm polyester-batting, 2 for the set

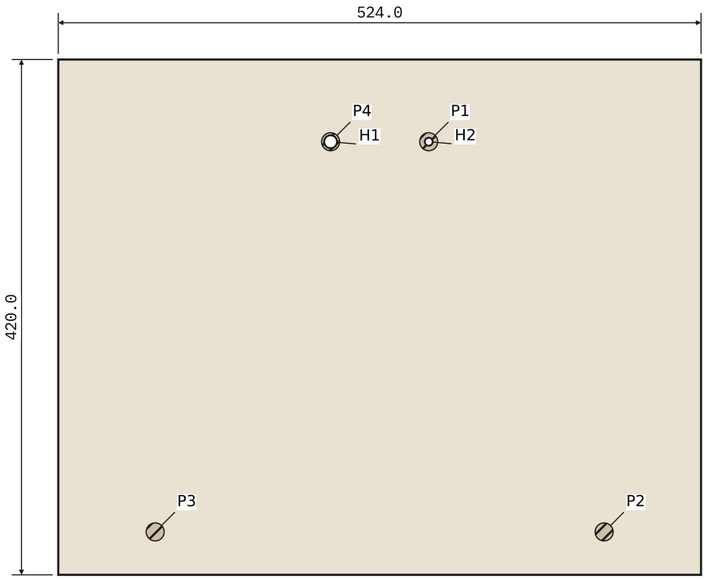
Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.



top: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø14.7	302.0	353.0	15 from face
P2	Ø14.7	445.0	35.0	15 from face
P3	Ø14.7	79.0	35.0	15 from face
P4	Ø14.7	222.0	353.0	15 from face
H1	Ø10.5	222.0	353.0	through
H2	Ø6.5	302.0	353.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

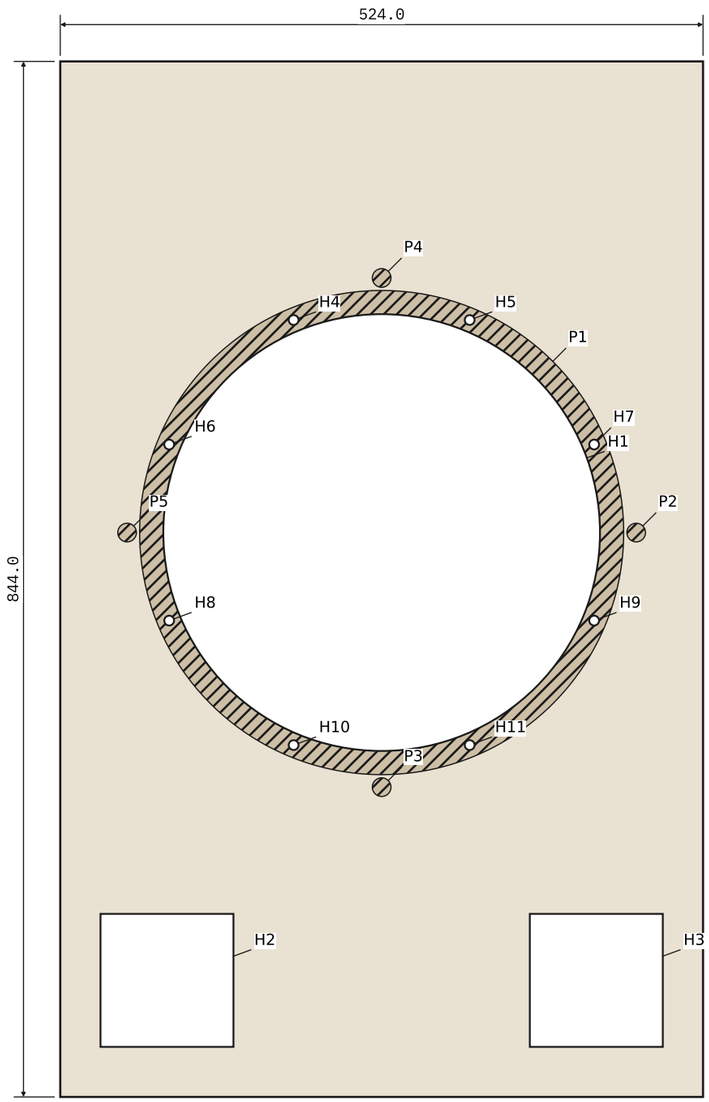
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	262.0	460.0	11 from face
P2	Ø15.2	469.5	460.0	3.2 from face
P3	Ø15.2	262.0	252.5	3.2 from face
P4	Ø15.2	262.0	667.5	3.2 from face
P5	Ø15.2	54.5	460.0	3.2 from face
H1	Ø356.0	262.0	460.0	through
H2	108.4 x 108.4	87.0	95.0	through
H3	108.4 x 108.4	437.0	95.0	through
H4	Ø8.0	190.2	633.2	through
H5	Ø8.0	333.8	633.2	through
H6	Ø8.0	88.8	531.8	through
H7	Ø8.0	435.2	531.8	through
H8	Ø8.0	88.8	388.2	through
H9	Ø8.0	435.2	388.2	through
H10	Ø8.0	190.2	286.8	through
H11	Ø8.0	333.8	286.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

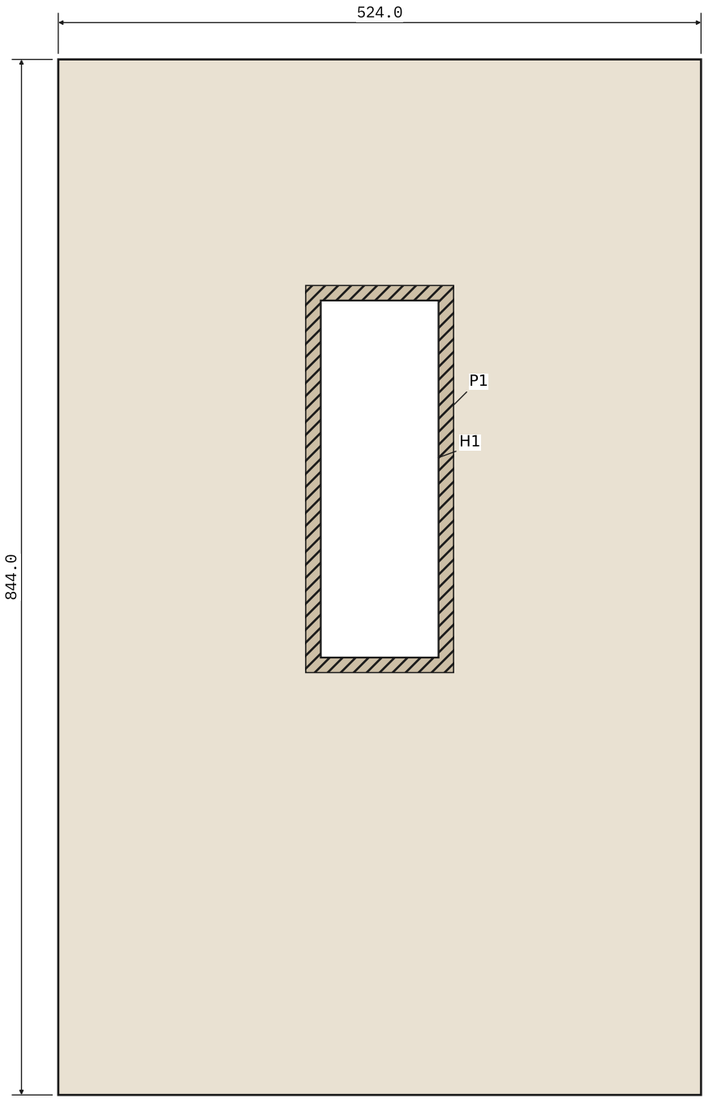
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	262.0	502.0	4.5 from face
H1	96.0 x 291.0	262.0	502.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

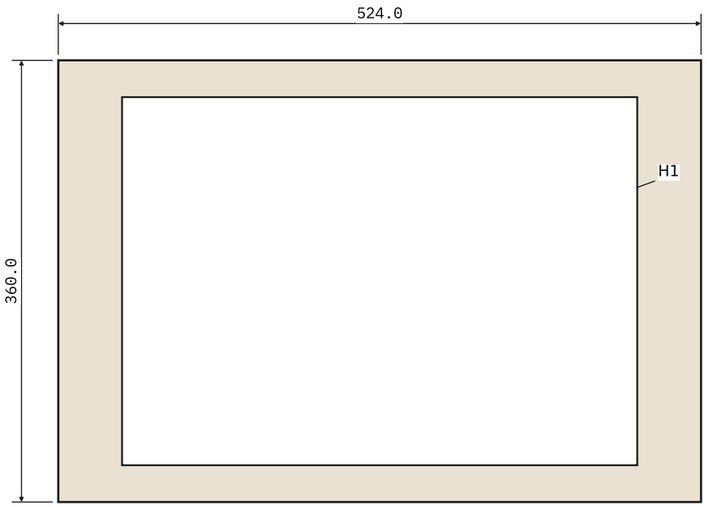
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	262.0	180.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

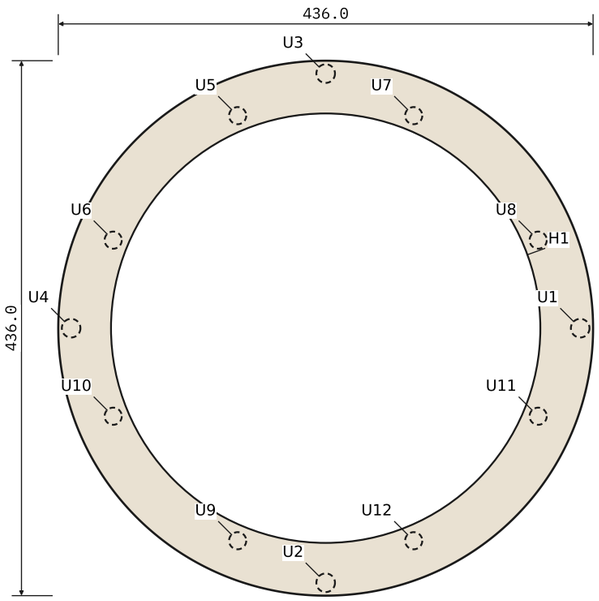
Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

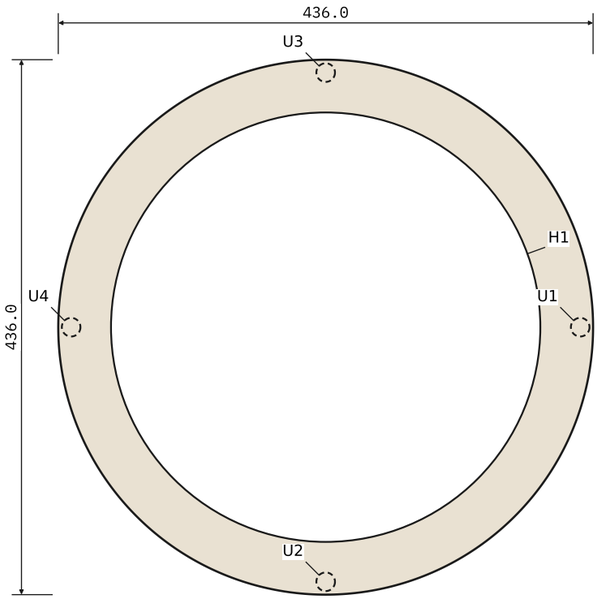
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

disc-frame: 12 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

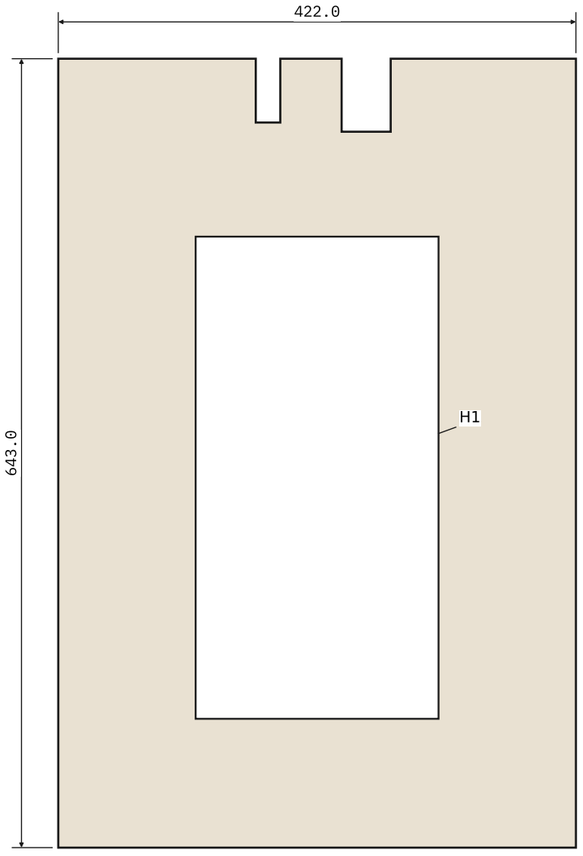
Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/lining-back-2.dxf. Marks listed at the right.



lining-back-2: 50 mm polyester-batting, 2 for the set, seen from its outer face

mark	size	x	y	depth
H1	198.0 x 393.0	211.0	301.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: lining-back-2

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

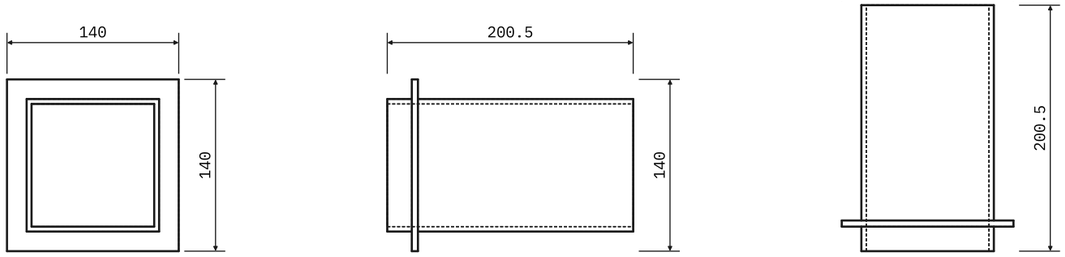
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. port-left, port-right: asa print, on its flange; 373 cm3 each; 4 for the set; 1:5

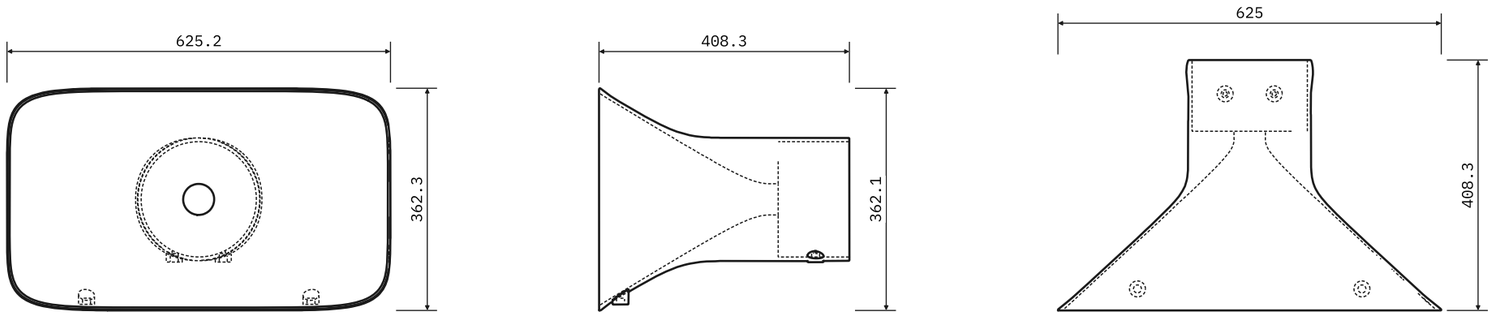
100 square, 200 long from the panel's face: the 2 together tune the box to 38 Hz; print 15 longer and trim to tune; fitted from inside, its 140 square flange on a 2 closed-cell PE foam seal on the panel's back, four wood screws through it



FRONT FROM THE RIGHT FROM ABOVE

11. waveguide: cast, cast-resin; 5222 cm3 each; 2 for the set; 1:10

mouth 610 x 346, 278 deep, 2 in throat; a 10 rim rounded right round, 625 x 362 outside; a shell 6 thick following its flare; the driver in a pod 200 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted clear resin in a two-part silicone mould from a CNC-cut plug, its faces polished satin (a slow deep-pour resin: the neck round the throat is solid, 5.2 L in all), or pigmented the same way for the opaque and marbled finishes



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

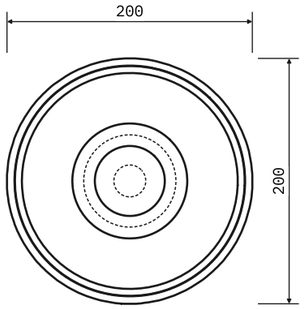
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

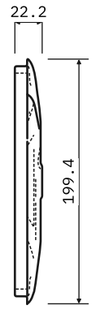
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. waveguide-cap: cast, cast-resin; 187 cm3 each; 2 for the set; 1:5

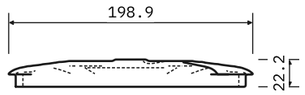
closes the pod behind the driver; a spigot locates it, three M4 screws hold it; the HF pair leaves through a post below; unplugged here, the horn lifts off its sockets



FRONT



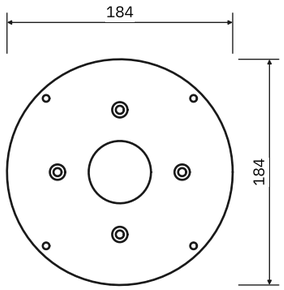
FROM THE RIGHT



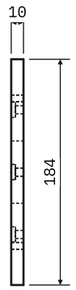
FROM ABOVE

14. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 x 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

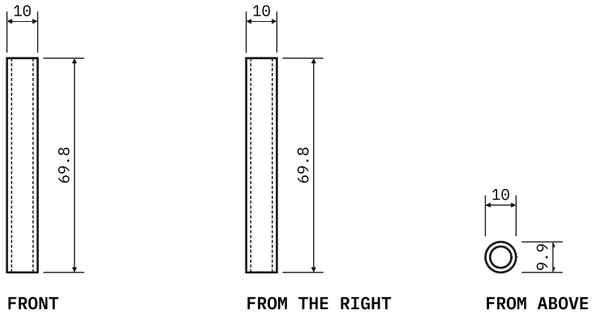
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

15. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

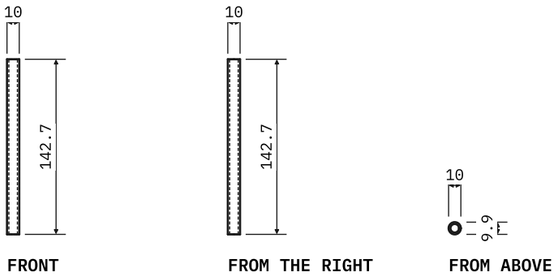
70 long (49 showing): stainless tube 10 x 7, threaded M10 x 1 8 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred



FRONT FROM THE RIGHT FROM ABOVE

16. post-3, post-4: lathe, stainless; 6 cm3 each; 4 for the set; 1:5

143 long (118 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred; the HF lead runs down inside it



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

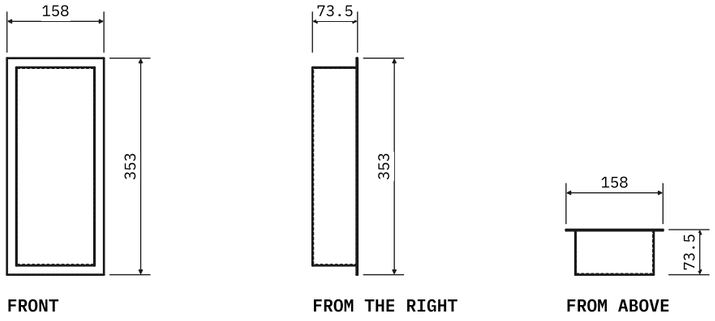
mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

21. amp-box: sheet metal, aluminium; 180 cm3 each; 2 for the set; 1:10

125 x 320 inside, 75 deep from the back: 1.5 aluminium folded, its corners welded, black anodised; a 15 flange round its open side, on a 3 EPDM gasket on the back's inner face, ten screws through it, so the box stays airtight and the mains electronics are enclosed in metal; two M12 cable glands in its front; re-trim the ports for the volume it takes



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	420 x 880	26.80
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
4	front-panel	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 360	13.68
7	woofer-ring	cut	2	6 mm MDF	436 x 436	4.28
8	port-left, port-right	print	4	ASA, FDM (outdoor, paintable)	373 cm3	34.13
9	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
10	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
11	waveguide	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	5222 cm3, 635 x 418 x 372	684.78
12	waveguide-cap	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	187 cm3, 210 x 32 x 210	138.21
13	hf	buy	2	B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat)	154 across, 86 deep, 4 x M6 on 102	512.00
14	hf-adaptor	lathe	2	Aluminium 6061	194 x 20 x 194	43.82
15	post-1, post-2	lathe	4	Stainless steel 304	20 x 20 x 80	15.96
16	post-3, post-4	lathe	4	Stainless steel 304	20 x 20 x 153	16.83
17	disc-frame	cut	2	12 mm MDF	436 x 436	4.28
18	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour		[PRICE]
19	cloth-disc-magnets	buy	2	The ring's and the disc's magnets: eight 15 x 3 neodymium discs (N42) and four 15 x 3 mild-steel discs, a set: four magnets in the panel and the steel discs in the ring's back hold the ring, four in the disc's frame hold the disc to the same		[PRICE]
20	bark-face, bark	buy	6	Bark for laminating, a square metre: paper birch (Betula papyrifera) sheet from fallen or felled trees, or a smooth-barked gum's shed bark pressed flat (or a UV-printed veneer of it); approve a sample board first		[PRICE]
21	amp-box	sheet metal	2	Aluminium 6061	305 x 500 x 2	85.72
22	amp-box-gasket	buy	2	EPDM sponge gasket, 3 mm, cut as a frame 15 wide for the amplifier compartment's flange		[PRICE]

24 more lines: bom.md

Notes

1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the front panel (set back 24 mm), the back, then the top.
3. The woofer flush in its rebate on a gasket, on eight M6 T-nuts from inside; the ring over its frame, held by four magnets in the panel under steel discs in its back; each port fitted from inside, trimmed to tune (impedance minimum at fb), its flange screwed to the panel on a foam seal.
4. The waveguide stands on four posts, stainless lamp tube 10 x 7, each threaded M10 x 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), its plain foot standing in a polyurethane sleeve in a brass socket pressed into the top, so the horn and its posts lift off as one. An M6 rod down the free rear post, from a thumb nut inside the pod to an insert under the top, holds them down. The HF lead, a round silicone cable 5 across, runs from the amplifier's compartment up the back, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone and the lead unplugged.

Parts list and notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-14 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15